

Keywords: video generation • visual pretraining • diffusion models • perception • general-purpose vision

GenCepcion

Google DeepMind Repurposes a Video Generative Diffusion Backbone as a Unified Perception Model, Achieving State-of-the-Art Across Five Vision Tasks with 7–500× Less Training Data

By Letian Wang et al. (Google DeepMind)

arXiv:2607.09024 • July 13, 2026

Computer vision has long operated on a modular paradigm: a separate specialist model for each perception task, each pretrained on its own curated dataset. Google DeepMind’s GenCepcion challenges this architecture at its root, demonstrating that the spatiotemporal priors baked into a large-scale text-to-video generative model transfer so effectively to perception that a single adapted backbone matches or exceeds task-specific specialists across depth, normals, pose, segmentation, and keypoint prediction — while requiring a small fraction of their training data.

AT A GLANCE

Problem: Vision perception requires separate specialist models per task; no single backbone generalizes across modalities with competitive performance.

Why it matters: A unified perception backbone would reduce maintenance overhead and could exploit shared geometric priors across all tasks simultaneously.

Method: Pre-trained text-to-video diffusion encoder adapted as a feed-forward perception model; task routing via text instructions; lightweight convolutional decoder per task.

Result: State-of-the-art on depth, surface normals, camera pose, segmentation (AP@50 = 61.3), and 3D keypoints (PCK@0.1 = 78.9%).

Evidence: 7–500× data efficiency vs. DepthAnything3, SAM3, VGGT-Omega, Sapiens; emergent synthetic-to-real generalization.

The Big Picture

The dominant paradigm in computer vision treats each perception task as an independent problem: depth estimation, surface normal prediction, camera pose estimation, segmentation, and keypoint prediction each receive a specialist model pretrained on large task-specific datasets. While effective, this paradigm is expensive to maintain and forecloses the possibility of cross-task transfer of geometric knowledge.

GenCepcion proposes a radical simplification: the rich spatiotemporal representations learned by a large-scale video generation model — including implicit 3D structure, motion physics, and visual semantics — are precisely the priors that perception tasks require, and can be transferred through minimal adaptation. The key insight is that video generation, to be accurate, must encode depth, surface orientation, object boundaries, and keypoint identity — all the outputs that perception specialists predict — as implicit intermediate representations.

Why Existing Approaches Fall Short

Specialist models excel within their training distribution but fail to leverage shared geometric structure across tasks. A model trained on depth estimation does not automatically learn better surface normal prediction, even though normals and depth are geometrically coupled.

Alternative general-purpose pretraining paradigms — including Video MAE, V-JEPA, and image diffusion models — provide weaker spatial and geometric priors than video generation because they either lack the generative objective (masking-based methods) or operate in 2D rather than the spatiotemporal domain (image diffusion). Video generation requires the model to implicitly solve geometry, physics, and scene understanding at every frame and across temporal transitions, encoding a uniquely rich inductive bias for perception.

Core Mechanism

GenCepcion extracts the denoising encoder of a pre-trained video diffusion model and adapts it to feed-forward perception through three modifications. First, a **patchification adapter** converts a single image into a sequence of spatial tokens compatible with the video backbone’s temporal token representation. Second, **text-steered task routing** injects a task-description embedding via cross-attention at each backbone layer, directing the backbone’s feature extraction toward the relevant geometric cues for the requested output. Third, a **lightweight convolutional decoder** maps backbone features to the target output resolution and format (depth map, normal map, segmentation mask, or keypoint heatmap) without adding significant parameter count.

Technical Formulation

Let $x \in \mathbb{R}^{H \times W \times 3}$ be an input image and t a natural-language task description. The video backbone f_θ (frozen then fine-tuned) maps (x, t) to a feature map:

$$F = f_\theta(\text{Patch}(x), \text{Enc}(t))$$

where $\text{Patch}(\cdot)$ is the patchification adapter and $\text{Enc}(\cdot)$ is a frozen text encoder. The task decoder g_ϕ then produces

output $\hat{y}$:

$$\hat{y} = g_\phi(F)$$

Training minimizes a task-specific loss $\mathcal{L}(\hat{y}, y)$ (scale-invariant depth loss, angular loss for normals, binary cross-entropy for segmentation, heatmap MSE for keypoints) while the backbone and decoder are jointly fine-tuned. The patchification adapter is initialized from the video model’s frame embedding layer and fine-tuned to handle single-image input without temporal context.

Learning Procedure

1. **Backbone source:** A large-scale text-to-video diffusion model pre-trained at Google DeepMind on diverse video corpora is used as the backbone.
2. **Patchification adapter fine-tuning:** Only the adapter is updated for the first 1000 steps to align single-image tokens with the backbone’s temporal token distribution.
3. **Joint fine-tuning:** Backbone and decoder are jointly fine-tuned on small per-task labeled datasets (ranging from 1K to 50K examples per task, depending on the task).
4. **Task-unified training:** All tasks are trained jointly with shared backbone weights, with task identity carried entirely by the text prompt.

Experimental Findings

Task (metric)	GenCepion	Specialist SOTA
Depth (AbsRel ↓)	0.038	0.041
Normals (mean angle ↓)	5.2°	5.8°
Pose (RRA@15 ↑)	82.4%	80.1%
Seg. (AP@50 ↑)	61.3	59.7
Keypoints (PCK@0.1 ↑)	78.9%	76.5%

Specialist comparisons: depth vs. DepthAnything3; normals vs. Lotus-2; pose vs. VGGT-Omega; segmentation vs. SAM3; keypoints vs. Sapiens. Data efficiency is 50× for depth, 100× for normals, 200× for pose, 7× for segmentation, and 500× for keypoints.

Ablations and Interpretation

Backbone type: Replacing the video diffusion backbone with an image diffusion backbone trained on comparable data degrades performance on all tasks by 3–8 points, confirming that spatiotemporal video pretraining — not generative pretraining per se — is the source of the perceptual advantage.

Text routing vs. fixed heads: Ablating text-steered routing (replacing it with fixed task heads) reduces performance by 2–4 points on depth and normals and eliminates zero-shot generalization to task variants described in novel

language.

Emergent synthetic-to-real transfer: A model trained exclusively on synthetic human videos achieves competitive performance on real-world footage and on animals — categories entirely absent from training. This behavior is absent when training with image-only or image-diffusion backbones, implicating spatiotemporal video priors as the source of the generalization.

Frozen vs. fine-tuned backbone: Freezing the backbone and adapting only the decoder reduces performance by 8–12 points across all tasks, indicating that the backbone features must be reshuffled to serve the regression objective, not just linearly decoded.

Reference

Letian Wang et al. **Video Generation Models are General-Purpose Vision Learners.** arXiv:2607.09024, July 2026. <https://arxiv.org/abs/2607.09024>